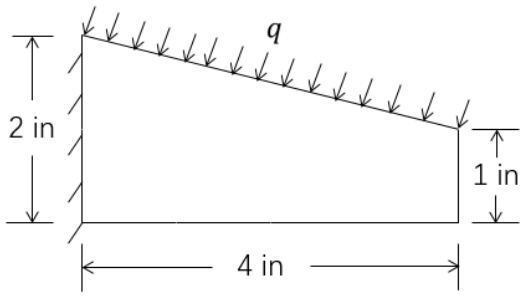
例题：托架应力分析

图示托架，一端固定，上表面法向压力$q=2\ \ lb/in^2$，托架尺寸如图，厚度$h=0.25in$ 。

材料弹性模量$E=10^4\,lb/in^2$，泊松比$v=0.2$



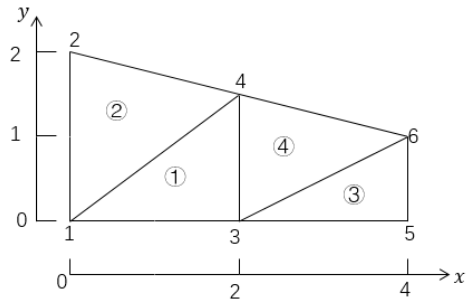
解：

该问题可按平面应力进行分析。

1.单元划分

整个结构划分4个三结点三角形单元,6 个结点，如图所示。

系统自由度数12（$u_1,v_1,u_2,v_2,u_3,v_3,u_4,v_4,u_5,v_5,u_6,v_6$）



2.单元分析（单元刚度方程）

单元1 刚度方程：

$h=0.25;E=10000;v=0.2$

平面应力本构矩阵$D=\dfrac{E}{1-v^2}\begin{bmatrix}1 & v & 0\\ v & 1 & 0\\ 0 & 0 & \dfrac{1-v}{2}\end{bmatrix}=\begin{bmatrix}10416.7 & 2083.33 & 0\\ 2083.33 & 10416.7 & 0\\ 0 & 0 & 4166.67\end{bmatrix}$

结点坐标

单元结点	整体结点编号	x	y
i	1	0	0
j	3	2	0
m	4	2	1.5

由结点坐标，

$x_i = 0;\qquad x_j = 2;\qquad x_m = 2$

$y_i = 0;\qquad y_j = 0;\qquad y_m = 1.5$

可求得

$b_i = -1.5;\qquad b_j = 1.5;\qquad b_m = 0$

$c_i = 0;\qquad c_j = -2;\qquad c_m = 2$

单元面积$A = \dfrac{a_i + a_j + a_m}{2} = 1.5$

$$[B] = \frac{1}{2A}\begin{bmatrix} b_i & 0 & b_j & 0 & b_m & 0 \\ 0 & c_i & 0 & c_j & 0 & c_m \\ c_i & b_i & c_j & b_j & c_m & b_m \end{bmatrix}$$

$$= \begin{bmatrix} -0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.666667 & 0 & 0.666667 \\ 0 & -0.5 & -0.666667 & 0.5 & 0.666667 & 0 \end{bmatrix}$$

单元刚度矩阵$\left[K^{(1)}\right] = hA[B]^T[D][B]$

$$= \begin{bmatrix} 976.563 & 0 & -976.563 & 260.471 & 0 & -260.417 \\ 0 & 390.625 & 520.833 & -390.625 & -520.833 & 0 \\ -976.563 & 520.833 & 1670.01 & -781.25 & -694.444 & 260.417 \\ 260.417 & -390.625 & -781.25 & 2126.74 & 520.833 & -1736.11 \\ 0 & -520.833 & -694.444 & 520.833 & 694.444 & 0 \\ -260.417 & 0 & 260.471 & -1736.11 & 0 & 1736.11 \end{bmatrix}$$

单元贡献自由度（1,2,5,6,7,8）

单元对总体结点载荷向量的贡献位置：$\begin{Bmatrix} 1 \\ 2 \\ 5 \\ 6 \\ 7 \\ 8 \end{Bmatrix}$

单元对系统刚度矩阵的贡献位置：$\begin{bmatrix} (1,1) & (1,2) & (1,5) & (1,6) & (1,7) & (1,8) \\ (2,1) & (2,2) & (2,5) & (2,6) & (2,7) & (2,8) \\ (5,1) & (5,2) & (5,5) & (5,6) & (5,7) & (5,8) \\ (6,1) & (6,2) & (6,5) & (6,6) & (6,7) & (6,8) \\ (7,1) & (7,2) & (7,5) & (7,6) & (7,7) & (7,8) \\ (8,1) & (8,2) & (8,5) & (8,6) & (8,7) & (8,8) \end{bmatrix}$

将单元方程加入整体方程，得：

$$\begin{bmatrix} 976.563 & 0 & 0 & 0 & -976.563 & 260.417 & 0 & -260.417 & 0 & 0 & 0 & 0 \\ 0 & 390.625 & 0 & 0 & 520.833 & -390.625 & -520.833 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -976.563 & 520.833 & 0 & 0 & 1671.07 & -781.25 & -694.444 & 260.417 & 0 & 0 & 0 & 0 \\ 260.417 & -390.625 & 0 & 0 & -781.25 & 2126.74 & 520.833 & -1736.11 & 0 & 0 & 0 & 0 \\ 0 & -520.833 & 0 & 0 & -694.444 & 520.833 & 694.444 & 0 & 0 & 0 & 0 & 0 \\ -260.417 & 0 & 0 & 0 & 260.417 & -1736.11 & 0 & 1736.11 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

单元2 刚度方程：

$h = 0.25; E = 10000; v = 0.2$

平面应力本构矩阵 $[D] = \begin{bmatrix} 10416.7 & 2083.33 & 0 \\ 2083.33 & 10416.7 & 0 \\ 0 & 0 & 4166.67 \end{bmatrix}$

结点坐标：

单元结点	整体结点编号	x	y
i	4	2	1.5
j	2	0	2
m	1	0	0

由单元结点坐标，可算出

$b_i = 2; \quad b_j = -1.5; \quad b_m = -0.5$

$c_i = 0; \quad c_j = 2; \quad c_m = -2$

单元面积A=2

$$[B] = \begin{bmatrix} 0.5 & 0 & -0.375 & 0 & -0.125 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & -0.5 \\ 0 & 0.5 & 0.5 & -0.375 & -0.5 & -0.125 \end{bmatrix}$$

单元刚度矩阵 $\left[K^{(2)}\right] = hA[B]^T[D][B]$

$$= \begin{bmatrix} 1302.08 & 0 & -976.563 & 260.417 & -325.521 & -260.417 \\ 0 & 520.833 & 520.833 & -390.625 & -520.833 & -130.208 \\ -976.563 & 520.833 & 1253.26 & -585.938 & -276.693 & 65.1042 \\ 260.417 & -390.625 & -585.938 & 1595.05 & 325.521 & -1204.43 \\ -325.521 & -520.833 & -276.693 & 325.521 & 602.214 & 195.313 \\ -260.417 & -130.208 & 65.1042 & -1204.43 & 195.313 & 1334.64 \end{bmatrix}$$

单元等效结点载荷向量（边界4 - 2），给定载荷分量$q_n = -20; q_t = 0$

结点坐标（2,1.5）-（0,2）$\Rightarrow$ 边长$L_{42} = 2.06155$

受载边界单位外法线分量$n_x = 0.242536, n_y = 0.970143.$

使用这些数据，可得：

$r_q^T = \{-1.25 \quad -5 \quad -1.25 \quad -5 \quad 0 \quad 0\}$

单元贡献自由度（7,8,3,4,1,2）

单元对总体载荷向量的贡献位置：$\begin{Bmatrix} 7 \\ 8 \\ 3 \\ 4 \\ 1 \\ 2 \end{Bmatrix}$

单元对系统刚度矩阵的贡献位置：$\begin{bmatrix} (7,7) & (7,8) & (7,3) & (7,4) & (7,1) & (7,2) \\ (8,7) & (8,8) & (8,3) & (8,4) & (8,1) & (8,2) \\ (3,7) & (3,8) & (3,3) & (3,4) & (3,1) & (3,2) \\ (4,7) & (4,8) & (4,3) & (4,4) & (4,1) & (4,2) \\ (1,7) & (1,8) & (1,3) & (1,4) & (1,1) & (1,2) \\ (2,7) & (2,8) & (2,3) & (2,4) & (2,1) & (2,2) \end{bmatrix}$

将单元方程加入整体方程，得：

$$\begin{bmatrix} 1578.78 & 195.313 & -276.693 & 325.521 & -976.563 & 260.417 & -325.521 & -781.25 & 0 & 0 & 0 & 0 \\ 195.313 & 1725.26 & 65.1042 & -1204.43 & 520.833 & -390.625 & -781.25 & -130.208 & 0 & 0 & 0 & 0 \\ -276.693 & 65.1042 & 1253.26 & -585.938 & 0 & 0 & -976.563 & 520.833 & 0 & 0 & 0 & 0 \\ 325.521 & -1204.43 & -585.938 & 1595.05 & 0 & 0 & 260.417 & -390.625 & 0 & 0 & 0 & 0 \\ -976.563 & 520.833 & 0 & 0 & 1671.01 & -781.25 & -694.444 & 260.417 & 0 & 0 & 0 & 0 \\ 260.417 & -390.625 & 0 & 0 & -781.25 & 2126.74 & 520.833 & -1736.11 & 0 & 0 & 0 & 0 \\ -325.521 & -781.25 & -976.563 & 260.417 & -694.444 & 520.833 & 1996.53 & 0 & 0 & 0 & 0 & 0 \\ -781.25 & -130.208 & 520.833 & -390.625 & 260.417 & -1736.11 & 0 & 2256.94 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ -1.25 \\ -5 \\ 0 \\ 0 \\ -1.25 \\ -5 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

同理，可得其余单元刚度矩阵和等效结点载荷。将所有单元加入总体方程，得系统总体刚度方程

$$\begin{bmatrix}
1578.78 & 195.313 & -276.693 & 325.521 & -976.563 & 260.417 & -325.521 & -781.25 & 0 & 0 & 0 & 0 \\
195.313 & 1725.26 & 65.1042 & -1204.43 & 520.833 & -390.625 & -781.25 & -130.208 & 0 & 0 & 0 & 0 \\
-276.693 & 65.1042 & 1253.26 & -585.938 & 0 & 0 & -976.563 & 520.833 & 0 & 0 & 0 & 0 \\
325.521 & -1204.43 & -585.938 & 1595.05 & 0 & 0 & 260.417 & -390.625 & 0 & 0 & 0 & 0 \\
-976.563 & 520.833 & 0 & 0 & 3125 & -520.833 & -1171.88 & 520.833 & -651.042 & 260.417 & -325.521 & -781.25 \\
260.417 & -390.625 & 0 & 0 & -520.833 & 4166.67 & 520.833 & -3385.42 & 520.833 & -260.417 & -781.25 & -130.208 \\
-325.521 & -781.25 & -976.563 & 260.417 & -1171.88 & 520.833 & 3125 & -520.833 & 0 & 0 & -651.042 & 520.833 \\
-781.25 & -130.208 & 520.833 & -390.625 & 520.833 & -3385.42 & -520.833 & 4166.67 & 0 & 0 & 260.417 & -260.417 \\
0 & 0 & 0 & 0 & -651.042 & 520.833 & 0 & 0 & 1692.71 & -781.25 & -1041.67 & 260.417 \\
0 & 0 & 0 & 0 & 260.417 & -260.417 & 0 & 0 & -781.25 & 2864.58 & 520.833 & -2604.17 \\
0 & 0 & 0 & 0 & -325.521 & -781.25 & -651.042 & 260.417 & -1041.67 & 520.833 & 2018.23 & 0 \\
0 & 0 & 0 & 0 & -781.25 & -130.208 & 520.833 & -260.417 & 260.417 & -2604.17 & 0 & 2994.79
\end{bmatrix}
\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix}
=
\begin{Bmatrix} 0 \\ 0 \\ -1.25+0 \\ -5+0 \\ 0 \\ 0 \\ -2.5 \\ -10 \\ 0 \\ 0 \\ -1.25 \\ -5 \end{Bmatrix}$$

边界位移约束条件：

由$u_1 = v_1 = u_2 = v_2 = 0$，刚度方程成为

$$\begin{bmatrix}
1578.78 & 195.313 & -276.693 & 325.521 & -976.563 & 260.417 & -325.521 & -781.25 & 0 & 0 & 0 & 0 \\
195.313 & 1725.26 & 65.1042 & -1204.43 & 520.833 & -390.625 & -781.25 & -130.208 & 0 & 0 & 0 & 0 \\
-276.693 & 65.1042 & 1253.26 & -585.938 & 0 & 0 & -976.563 & 520.833 & 0 & 0 & 0 & 0 \\
325.521 & -1204.43 & -585.938 & 1595.05 & 0 & 0 & 260.417 & -390.625 & 0 & 0 & 0 & 0 \\
-976.563 & 520.833 & 0 & 0 & 3125 & -520.833 & -1171.88 & 520.833 & -651.042 & 260.417 & -325.521 & -781.25 \\
260.417 & -390.625 & 0 & 0 & -520.833 & 4166.67 & 520.833 & -3385.42 & 520.833 & -260.417 & -781.25 & -130.208 \\
-325.521 & -781.25 & -976.563 & 260.417 & -1171.88 & 520.833 & 3125 & -520.833 & 0 & 0 & -651.042 & 520.833 \\
-781.25 & -130.208 & 520.833 & -390.625 & 520.833 & -3385.42 & -520.833 & 4166.67 & 0 & 0 & 260.417 & -260.417 \\
0 & 0 & 0 & 0 & -651.042 & 520.833 & 0 & 0 & 1692.71 & -781.25 & -1041.67 & 260.417 \\
0 & 0 & 0 & 0 & 260.417 & -260.417 & 0 & 0 & -781.25 & 2864.58 & 520.833 & -2604.17 \\
0 & 0 & 0 & 0 & -325.521 & -781.25 & -651.042 & 260.417 & -1041.67 & 520.833 & 2018.23 & 0 \\
0 & 0 & 0 & 0 & -781.25 & -130.208 & 520.833 & -260.417 & 260.417 & -2604.17 & 0 & 2994.79
\end{bmatrix}
\begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix}
=
\begin{Bmatrix} F_{1x}^* \\ F_{1y}^* \\ -1.25+F_{2x}^* \\ -5+F_{2y}^* \\ 0 \\ 0 \\ -2.5 \\ -10 \\ 0 \\ 0 \\ -1.25 \\ -5 \end{Bmatrix}$$

降阶方程：

$$\begin{bmatrix}
3125 & -520.833 & -1171.88 & 520.833 & -651.042 & 260.417 & -325.521 & -781.25 \\
-520.833 & 4166.67 & 520.833 & -3385.42 & 520.833 & -260.417 & -781.25 & -130.208 \\
-1171.88 & 520.833 & 3125 & -520.833 & 0 & 0 & -651.042 & 520.833 \\
520.833 & -3385.42 & -520.833 & 4166.67 & 0 & 0 & 260.417 & -260.417 \\
-651.042 & 520.833 & 0 & 0 & 1692.71 & -781.25 & -1041.67 & 260.417 \\
260.417 & -260.417 & 0 & 0 & -781.25 & 2864.58 & 520.833 & -2604.17 \\
-325.521 & -781.25 & -651.042 & 260.417 & 1041.67 & 520.833 & 2018.23 & 0 \\
-781.25 & -130.208 & 520.833 & -260.417 & 260.417 & -2604.17 & 0 & 2994.79
\end{bmatrix}
\begin{Bmatrix} u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix}
=
\begin{Bmatrix} 0 \\ 0 \\ -2.5 \\ -10 \\ 0 \\ 0 \\ -1.25 \\ -5 \end{Bmatrix}$$

解方程得：

$$u_3 = -0.0103553, v_3 = -0.0255297, u_4 = 0.00472765, v_4 = -0.0247357$$

$$u_5 = -0.0131394, v_5 = -0.0554931, u_6 = 0.0000838902, v_6 = -0.0555664$$

约束反力计算：

在总体方程中取约束反力对应的前4 个方程，

$$\begin{bmatrix} 1578.78 & 195.313 & -276.693 & 325.521 & -976.563 & 260.417 & -325.521 & -781.25 & 0 & 0 & 0 & 0 \\ 195.313 & 1725.26 & 65.1042 & -1204.43 & 520.833 & -390.625 & -781.25 & -130.28 & 0 & 0 & 0 & 0 \\ -276.693 & 65.1042 & 1253.26 & -585.938 & 0 & 0 & -976.563 & 520.833 & 0 & 0 & 0 & 0 \\ 325.521 & -1204.43 & -585.938 & 1595.05 & 0 & 0 & 260.417 & -390.625 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{Bmatrix} = \begin{Bmatrix} F_{1x}^* + 0 \\ F_{1y}^* + 0 \\ F_{2x}^* - 1.25 \\ F_{2y}^* - 5 \end{Bmatrix}$$

代入结点位移，整理后：

$$\begin{Bmatrix} F_{1x}^* \\ F_{1y}^* \\ F_{2x}^* \\ F_{2y}^* \end{Bmatrix} = \begin{bmatrix} 1578.78 & 195.313 & -276.693 & 325.521 & -976.563 & 260.417 & -325.521 & -781.25 & 0 & 0 & 0 & 0 \\ 195.313 & 1725.26 & 65.1042 & -1204.43 & 520.833 & -390.625 & -781.25 & -130.28 & 0 & 0 & 0 & 0 \\ -276.693 & 65.1042 & 1253.26 & -585.938 & 0 & 0 & -976.563 & 520.833 & 0 & 0 & 0 & 0 \\ 325.521 & -1204.43 & -585.938 & 1595.05 & 0 & 0 & 260.417 & -390.625 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -0.0103553 \\ -0.0255297 \\ 0.00472765 \\ -0.0247357 \\ -0.0131394 \\ -0.0554931 \\ 0.0000838902 \\ -0.0555664 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 0 \\ -1.25 \\ -5 \end{Bmatrix}$$

解得：

F_{1x}^*=21.25, $F_{1y}^* = 4.10648$, $F_{2x}^* = -16.25$, $F_{2y}^* = 15.8935$

反力求和，在x 方向上，$F_{1x}^* + F_{2x}^* = 5$ ；在y 方向上，$F_{1y}^* + F_{2y}^* = 20$

支反力应与外力平衡。

为确定总外载荷及其在x 和 y方向的分量，需$6-2$ 边长和法向量

$6-2$边端点坐标：

6点：$x_1 = 4, y_1 = 1$;　2点：$x_2 = 0, y_2 = 2$;

边长　$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = 4.12311$

法向量　$n_x = (y_2 - y_1)/L = 0.242536, n_y = -(x_2 - x_1)/L = 0.970143$

总载荷 $= qhL = -20.6155$

x 向分量 $= qhLn_x = -5$　　与支反力总和相等

y 向分量 $= qhLn_y = -20$

解答满足总体平衡